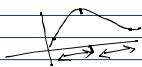


Measure of dispersion:

Measure of dispersion describe the spread or variability of a dataset. They indicate how much the values in a dataset from the central tendency.

Common Measures:-

- ① Range
- ② Variance
- ③ Standard deviation
- ④ Inter Quartile Range (IQR)



① Range

Def:- The difference between maximum value & min value

$$\text{Range} = \text{max Value} - \text{min Value}$$

$$\text{Age} = \{14, 13, 10, 20, 25, 25, 15\}$$

$$= 25 - 10$$

$$= 15 \Rightarrow 15 \text{ Range}$$

Char:-

- i) Simple to Calculate
- ii) Sensitive to outlier

② Variance

Variance Measure the avg squared deviation of each value from the Mean. It provides an sense of how much the values in a dataset vary.

Population Variance (σ^2)

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$

x_i → Data point
 μ → Population mean
 N → Population Size

Sample Variance (s^2)

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1} \rightarrow \text{Bessel's correction}$$

x_i → Data point
 $\bar{x}$ → Sample Mean
 n → Sample Size

Ex:- $\{5, 8, 12, 15, 20\} \Rightarrow \text{Variance}$

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$

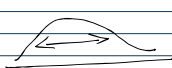
$$\mu = \frac{5+8+12+15+20}{5} = 12$$

$$= \frac{(5-12)^2 + (8-12)^2 + (12-12)^2 + (15-12)^2 + (20-12)^2}{5}$$

$$\text{Variance} = 22.6$$

Char:-

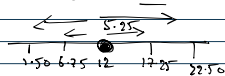
- ① It provides a precise measure of variability
- ② Squared of original units $\Rightarrow$ change
- ③ Sensitive to outliers



③ Standard deviation

The standard deviation is the square root of the Variance

$$\sigma = \sqrt{22.6} \approx 4.75$$



$$\{5, 8, 12, 15, 20\} \rightarrow 2 \text{ score}$$

Characteristics

- $\Rightarrow$ It provides a clear measure of spread in the same unit
- $\Rightarrow$ Sensitive to outliers

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